

Looping in Java

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Looping (also called iteration) is a programming concept that allows a block of code to execute repeatedly until a specified condition becomes false. Instead of writing the same statements multiple times, makes. Program shorter, easier to read, and more efficient.

Why do we use loops?

Loops are used when we need to:

- Display numbers repeatedly.
- Process element of an array or Collection.
- Read multiple inputs from the user.
- Perform a repetitive calculations.
- Generate patterns.
- Traverse files or data structures.

In Java three types of loops

Entry control loop

1. while loop

2. for loop

Exit control loop.

1. do { } while loop.

Collection loop [List, Dictionary, HashMap, etc.]

1. for each loop. Arrays
for (variable : collection object)

Syntax of while loop:-

```
while (<condition>) {
```

```
// body of the loop.
```

```
}
```

```
package Loops;  
import java.util.*;
```

```
/**  
 * Write a description of class WhileDemo here.  
 *  
 * @author (your name)  
 * @version (a version number or a date)  
 */  
public class WhileDemo  
{  
    public static void main(String[] args){  
        Scanner sc = new Scanner(System.in);
```

```

int start = 1, n;
System.out.print("\nEnter number of times: ");
n = sc.nextInt();
while(start <= n){
    System.out.print("\n" + start+": Hello!");
    start++;
}
}
}

```

For loop % [Skill: If we know number of iteration, then we use for loop],
 For loop is an advance form of while loop.

Syntax:

for (<initializations> ; <condition> ; <increment/decrement>) {

// loop body

}

e.g.

for (i = 1 ; i <= n ; i++) {

// this loop will execute for 'n' number of times

}

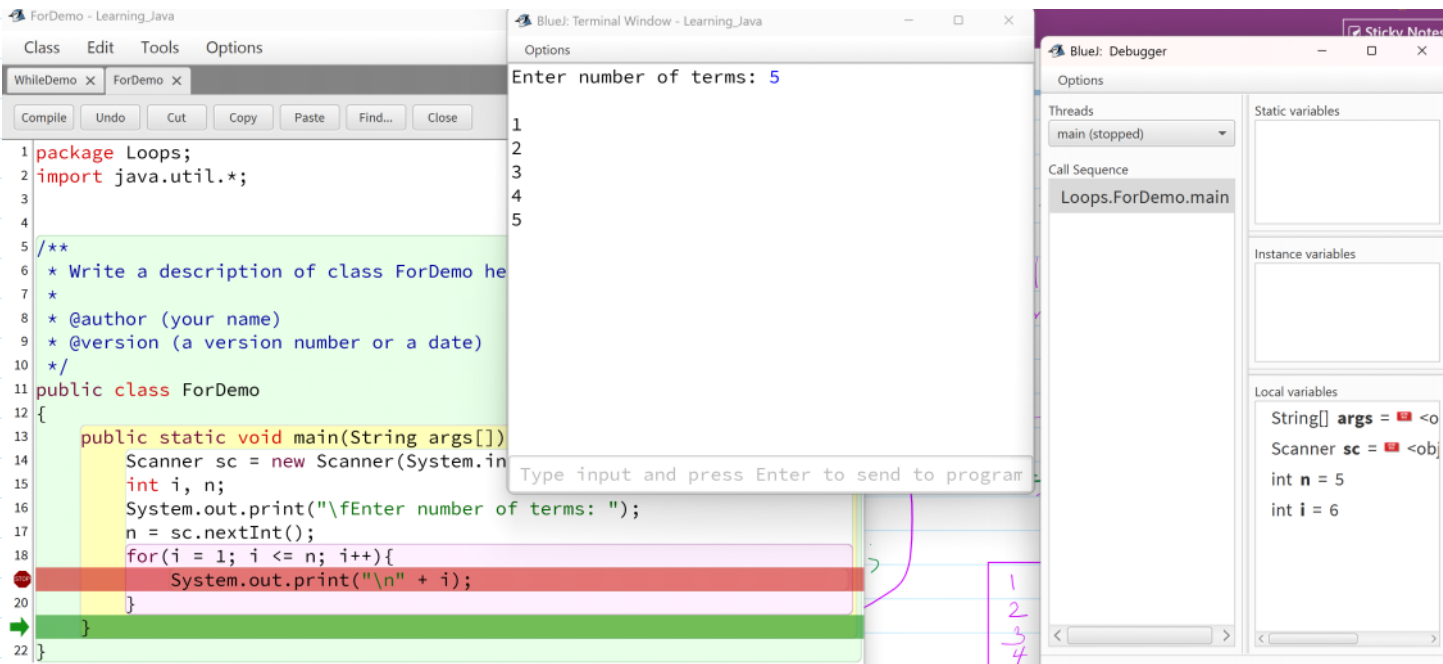
When the value of i exceeds the value of 'n' then for loop terminates.

F
 T
 for (i = 1 ; i <= n ; i++) {
 3 → System.out.print("n" + i);
 }
 →

n = 5

i = 1 2 3 4 5 6

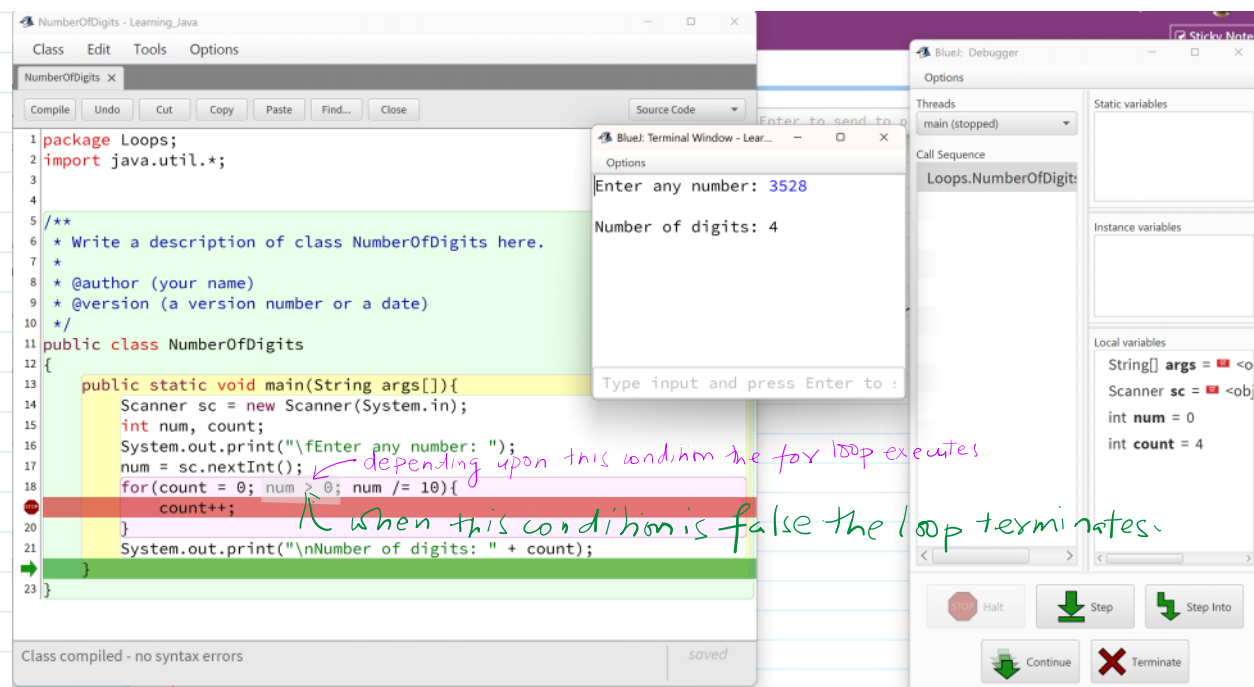
1
2
3
4
5



WAP to input a number and find out how many digits present in it.

Input: Enter any number: 3456

Output: Number of digits: 4



do...while() loop

Syntax:

```

do{
    //write your code to repeat
    ...
}while(<condition>);

```

This loop body always execute at least once, because the condition is tested after the execution of the body.

This type of loop is very much useful to control user input.

```
package Loops;

import javax.swing.*;
/**
 * Write a description of class DowhileDemo here.
 *
 * @author (your name)
 * @version (a version number or a date)
 */
public class DowhileDemo
{
    public static void main(String[] args){
        int n;
        do{
            String data = JOptionPane.showInputDialog("Enter a positive number",
"0");
            n = Integer.parseInt(data);
            if(n > 0){
                break;
            }
            else{
                JOptionPane.showConfirmDialog(null,
                    "\nKindly read the prompt in the input dialog box.");
            }
        }while(true);
        JOptionPane.showConfirmDialog(null, "This is valid input.", "Valid Input",
JOptionPane.OK_OPTION, JOptionPane.INFORMATION_MESSAGE);
    }
}
```

Output:

Given Inside dialog boxes.

Another example of do...while loop:

```
package Loops;

import javax.swing.*;
/**
 * Write a description of class PalindromeOrPerfect here.
 * Write a menu driven program to accept a number from the user.
 * Check and display whether it is a Palindrome or a Perfect number.
 * (a) Palindrome number: (A number is a Palindrome which when read in
 * the reverse order is the same as the original number)
 * Example: 11, 101, 151, etc.
 * (b) Perfect number: (A number is called Perfect if it
 * is equal to the sum of its
 * factors other than the number itself.)
 * Example: 6
 * 1 + 2 + 3
```

```

*
* @author (your name)
* @version (a version number or a date)
*/
public class PalindromeOrPerfect
{
    public static void main(String[] args){
        String choice, input, output="";
        String menu = "\nPress 1 for palindrome"+
            "\nPress 2 for perfect number"+
            "\nPress 3 for exit"+
            "\nEnter your choice: ";
        do{
            choice = JOptionPane.showInputDialog(menu, "1");
            switch(choice){
                case "1":
                    input = JOptionPane.showInputDialog(null,
                        "Enter any positive number: ");
                    int num = Integer.parseInt(input);
                    int rev = 0, t = num;
                    do{
                        rev = rev * 10 + t % 10;
                        t = t / 10;
                    }while(t > 0);
                    if(num == rev){
                        JOptionPane.showMessageDialog(null, "Palindrome",
                            "Check Palindrome", JOptionPane.OK_OPTION);
                    }
                    else{
                        JOptionPane.showMessageDialog(null, "Not Palindrome",
                            "Check Palindrome", JOptionPane.OK_OPTION);
                    }
                    break;
                case "2":
                    input = JOptionPane.showInputDialog(null,
                        "Enter any positive number: ");
                    num = Integer.parseInt(input);
                    int sum = 0, d = 1; t = num;
                    do{
                        if(t % d == 0){
                            sum += d;
                        }
                        d++;
                    }while(d <= t/2);
                    if(sum == num){
                        JOptionPane.showMessageDialog(null, "Perfect number",
                            "Check Perfect", JOptionPane.OK_OPTION);
                    }
                    else{
                        JOptionPane.showMessageDialog(null, "Not a Perfect Number",
                            "Check Perfect", JOptionPane.OK_OPTION);
                    }
                    break;
                case "3":

```

```
        JOptionPane.showMessageDialog(null, "Exiting...",
            "Exit", JOptionPane.OK_OPTION);
        System.exit(3);
    default:
        JOptionPane.showMessageDialog(null, "Wrong option Selected",
            "Menu message", JOptionPane.OK_OPTION);
    }
}while(true);
}
```